

CISCO NEXUS N5K-C5596UP

Physical Data Center Infrastructure Recovery

TFTP-Based NX-OS Recovery & Hardware Validation

Brandon Stevenson

Physical Cisco Nexus N5K-C5596UP switch

Serial-console recovery | TFTP image transfer | Bootflash management

Hardware diagnostics | Interface inspection | System initialization

19 documented phases | Original terminal command output

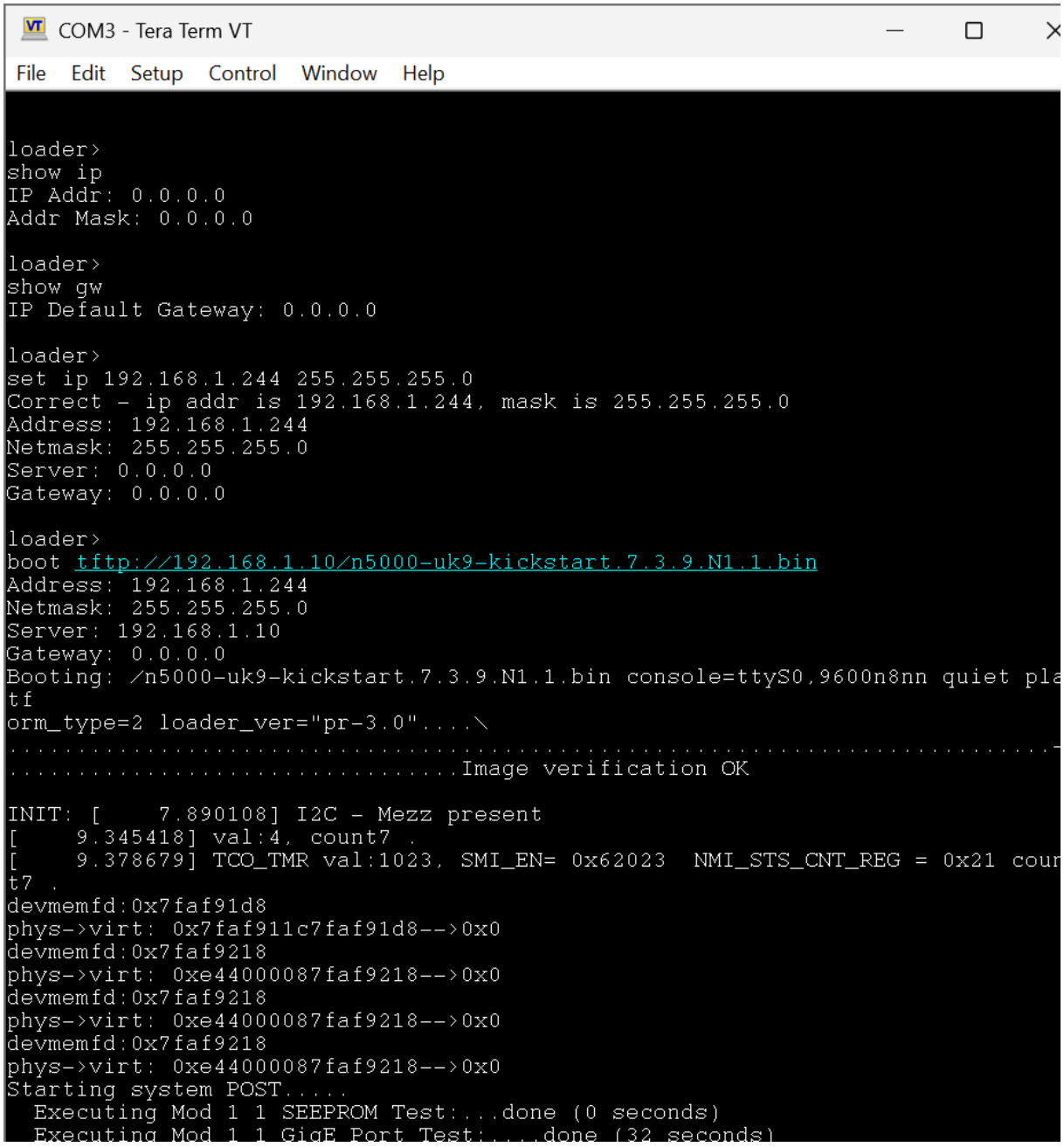
PHASE 1

Bootloader management and network boot

show ip / show gw
Checks the bootloader network settings before starting the transfer.

set ip 192.168.1.244 255.255.255.0
Assigns the switch a temporary management address on the same subnet as the TFTP server.

boot tftp://192.168.1.10/n5000-uk9-kickstart.7.3.9.N1.1.bin
Downloads and starts the kickstart image from the TFTP server. Image verification completes successfully.



Terminal capture 1 - Recorded by Brandon Stevenson

PHASE 2

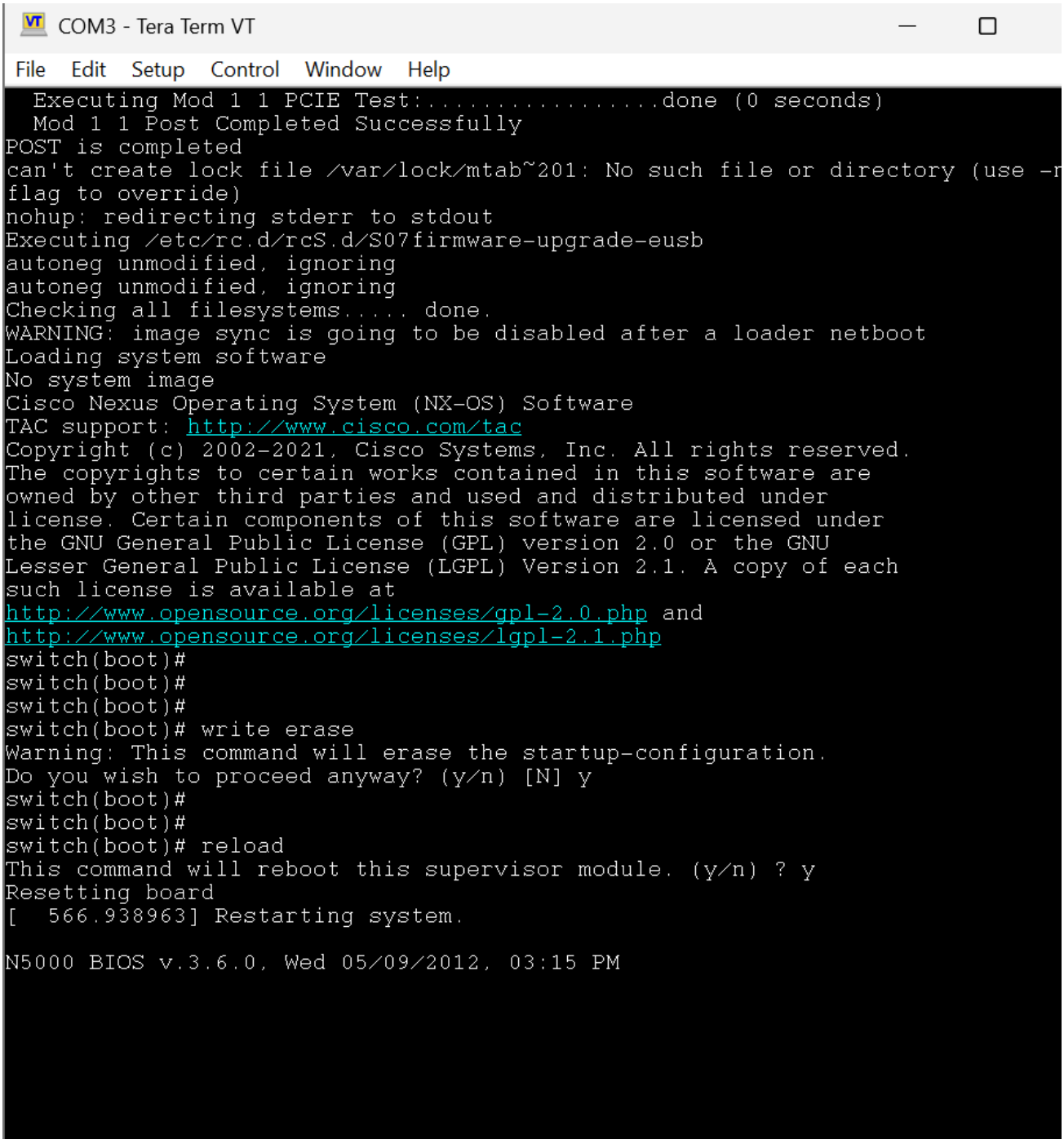
Boot into kickstart and clear startup settings

write erase

Removes the saved switch configuration so the next startup begins without the previous settings.

reload

Restarts the supervisor after the configuration reset.



Terminal capture 2 - Recorded by Brandon Stevenson

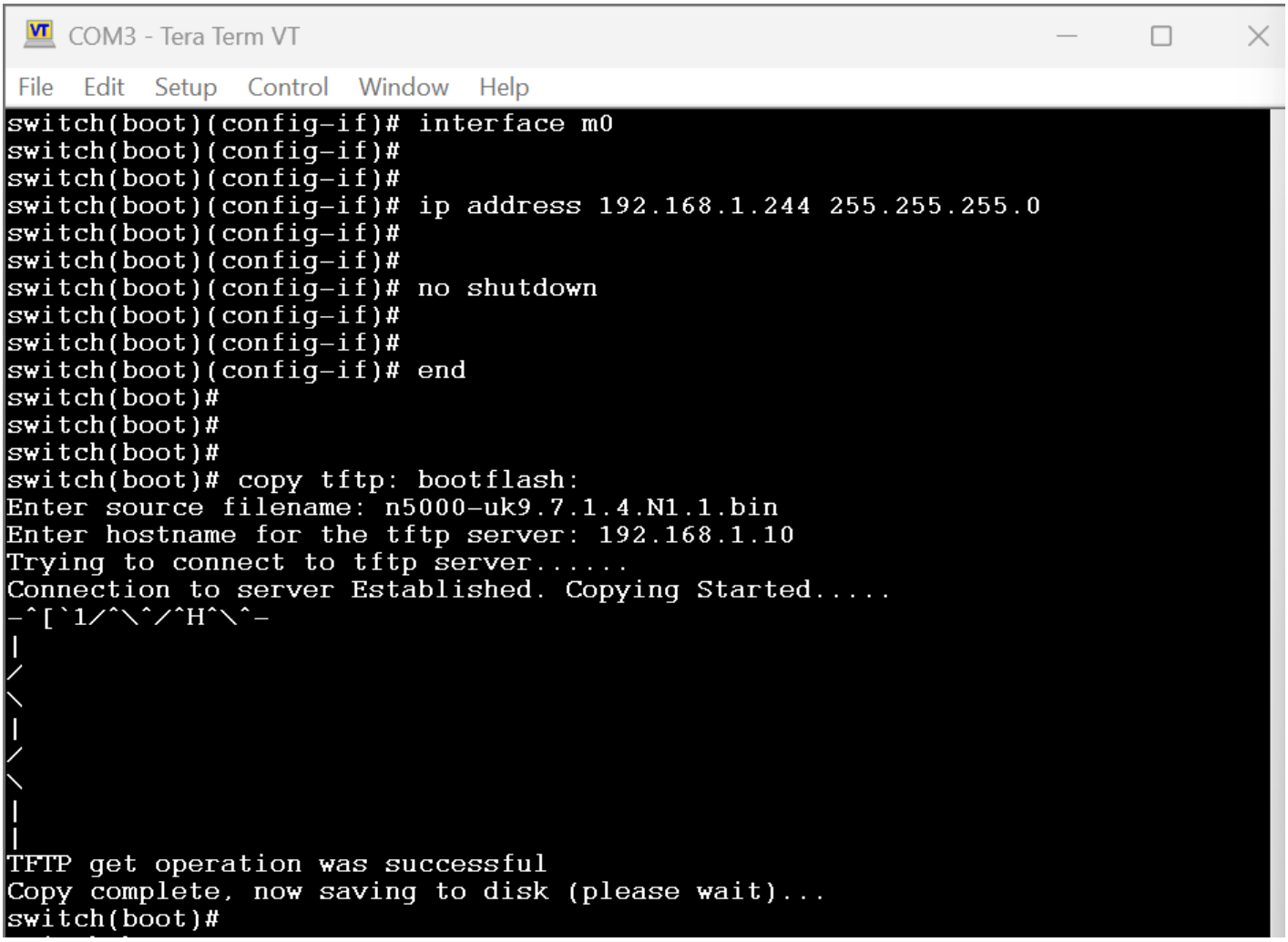
PHASE 3

Copy the NX-OS image from TFTP to bootflash

interface m0 / ip address 192.168.1.244 255.255.255.0
Configures the management interface within the kickstart environment.

no shutdown / end
Enables the interface and returns to the boot prompt.

copy tftp: bootflash:
Starts an interactive TFTP copy to internal bootflash. The transfer completes and the image is saved locally.



Terminal capture 3 - Recorded by Brandon Stevenson

PHASE 4

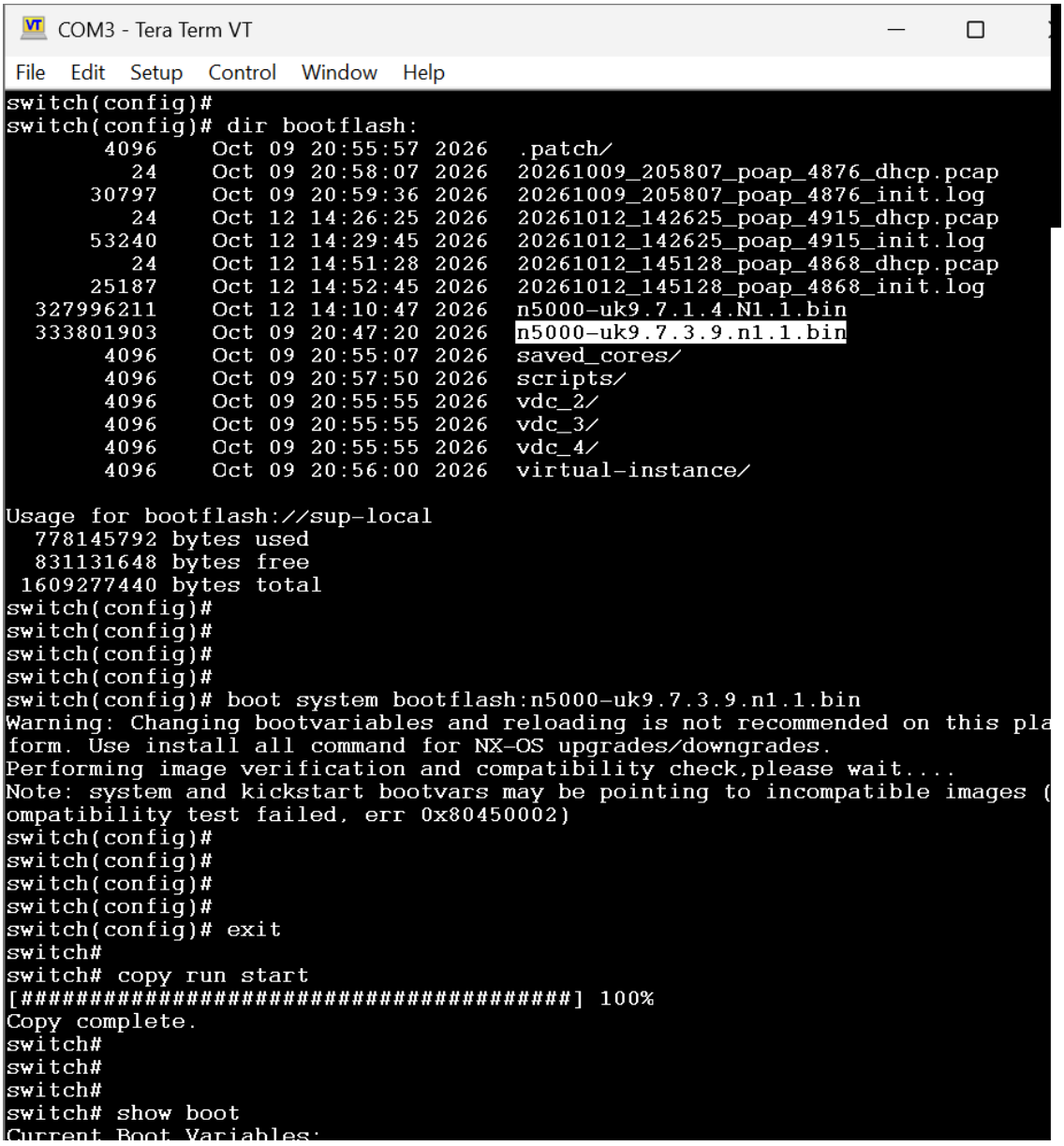
Inspect bootflash and set the system boot image

dir bootflash:
Lists the image files and other content stored on internal flash.

boot system bootflash:n5000-uk9.7.3.9.N1.1.bin
Attempts to set the system image for startup. NX-OS reports an image-pair compatibility warning.

copy run start
Saves the running configuration.

show boot
Displays the configured boot paths.



Terminal capture 4 - Recorded by Brandon Stevenson

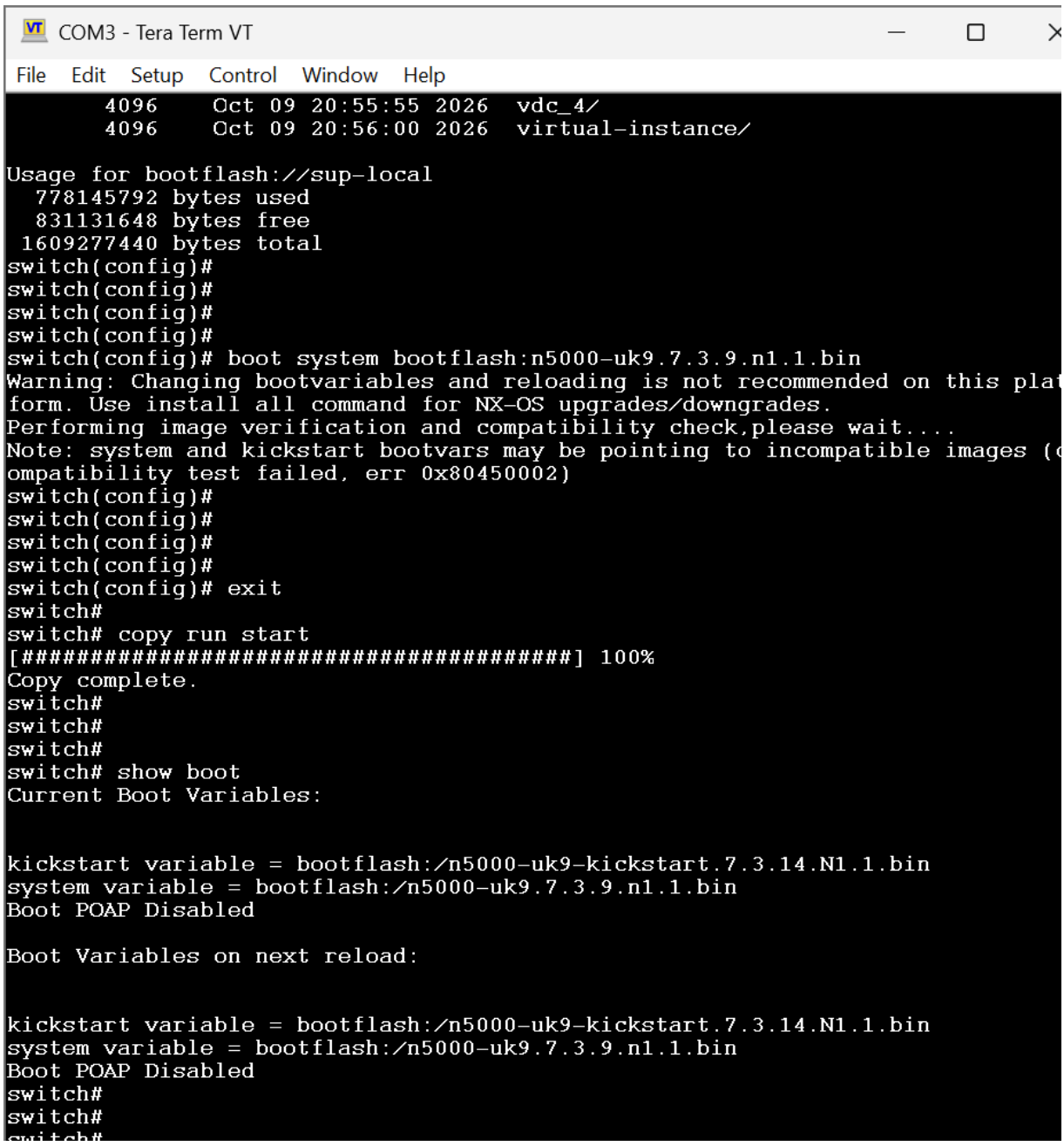
PHASE 5

Review kickstart and system boot variables

boot system bootflash:n5000-uk9.7.3.9.N1.1.bin
Attempts to set the system image. The compatibility check indicates the configured kickstart and system releases do not match.

copy run start
Saves the current configuration.

show boot
Displays both boot variables so the image versions can be compared.



Terminal capture 5 - Recorded by Brandon Stevenson

PHASE 6

Inspect local image files

dir bootflash:

Checks the copied NX-OS images and remaining storage content.

CLI parameter warning

The terminal reports invalid characters after the directory listing. The image files remain visible in the output.

```
switch(boot)#
switch(boot)# dir bootflash:
        4096  Oct 09 2026 20:55:57  .patch/
          24  Oct 09 2026 20:58:07  20261009_205807_poap_4876_dhcp.pcap
       30797  Oct 09 2026 20:59:36  20261009_205807_poap_4876_init.log
   327996211  Oct 12 2026 14:10:47  n5000-uk9.7.1.4.N1.1.bin
   333801903  Oct 09 2026 20:47:20  n5000-uk9.7.3.9.n1.1.bin
        4096  Oct 09 2026 20:55:07  saved_cores/
        4096  Oct 09 2026 20:57:50  scripts/
        4096  Oct 09 2026 20:55:55  vdc_2/
        4096  Oct 09 2026 20:55:55  vdc_3/
        4096  Oct 09 2026 20:55:55  vdc_4/
        4096  Oct 09 2026 20:56:00  virtual-instance/
Parameter contains one or more invalid characters (*$%&!~`)(<>|)
switch(boot)#
```

Terminal capture 6 - Recorded by Brandon Stevenson

PHASE 7

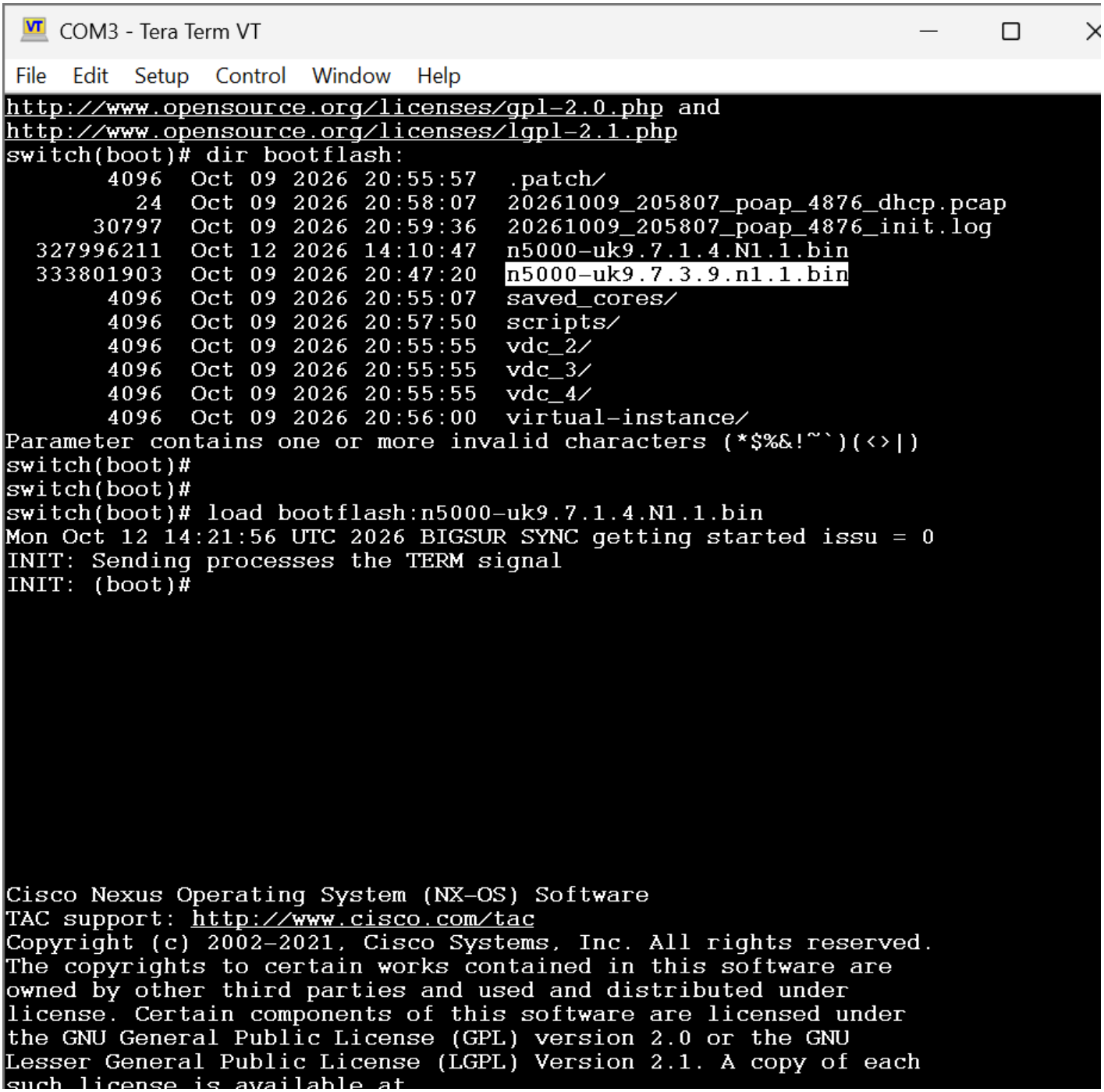
Load the local NX-OS system image

dir bootflash:

Confirms the system image filename on internal flash.

load bootflash:n5000-uk9.7.1.4.N1.1.bin

Loads the selected system image from bootflash and starts NX-OS.



Terminal capture 7 - Recorded by Brandon Stevenson

PHASE 8

Identify NX-OS and the switch platform

terminal length 0

Turns off CLI pagination so command output can be read without page breaks.

show version

Displays the operating-system release, active images, BIOS information, uptime, and Nexus chassis model.



Terminal capture 8 - Recorded by Brandon Stevenson

PHASE 9

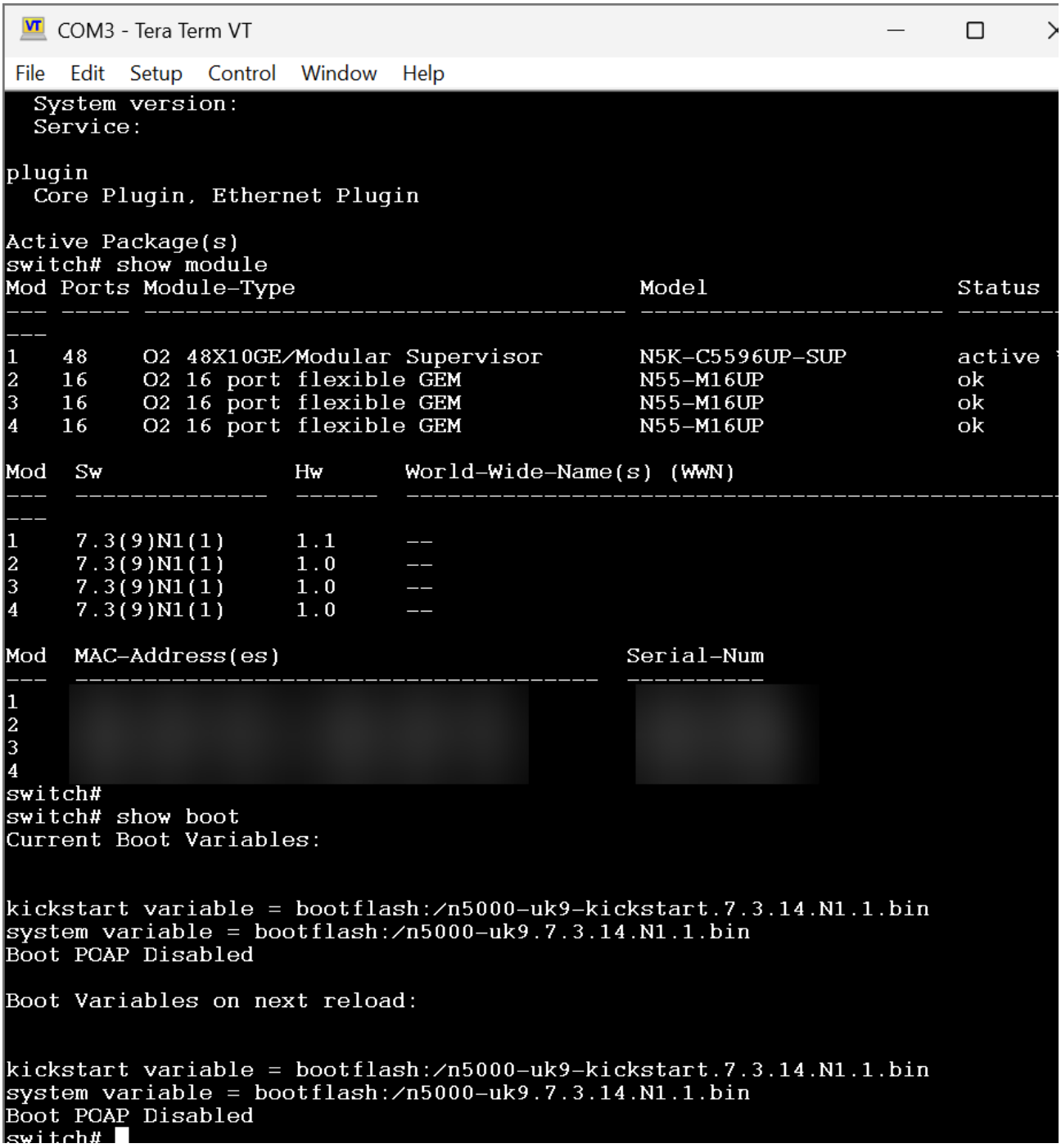
Inspect modules and boot paths

show module

Reports installed modules, model information, and operational status.

show boot

Displays the current and next-boot kickstart and system paths. The differing image versions are visible here.



Terminal capture 9 - Recorded by Brandon Stevenson

PHASE 10

Check chassis power status

show environment power

Reports power-supply state, allocated power, and redundancy. One power supply is operating; the other reports fail/shutdown.

COM3 - Tera Term VT

File Edit Setup Control Window Help

```
switch#
switch# show environment power


Power Supply:
Voltage: 12 Volts
--------------------------------------------------------
PS  Model                Input Power       Current   Status
                         Type  (Watts)     (Amps)
--------------------------------------------------------
1   NXA-PAC-1100W        AC    1080.00     90.00     ok
2   --                   --         --        --     fail/shutdown


Mod Model                   Power     Current     Power     Current     Stat
s
                            Requested Requested   Allocated Allocated

                            (Watts)   (Amps)      (Watts)   (Amps)

--- --------------------  -------   ----------  --------- ----------  ----
-----
1    N5K-C5596UP-SUP        648.00    54.00       648.00    54.00       powe
ed-up
2    N55-M16UP              90.00     7.50        90.00     7.50        powe
ed-up
3    N55-M16UP              90.00     7.50        90.00     7.50        powe
ed-up
4    N55-M16UP              90.00     7.50        90.00     7.50        powe
ed-up


Power Usage Summary:
--------------------
Power Supply redundancy mode:                 Redundant
Power Supply redundancy operational mode:     Non-redundant

Total Power Capacity                             1080.00 W

Power reserved for Supervisor(s)                  648.00 W
Power currently used by Modules                   270.00 W

                                                -------------
Total Power Available                             162.00 W
                                                -------------
switch#
```

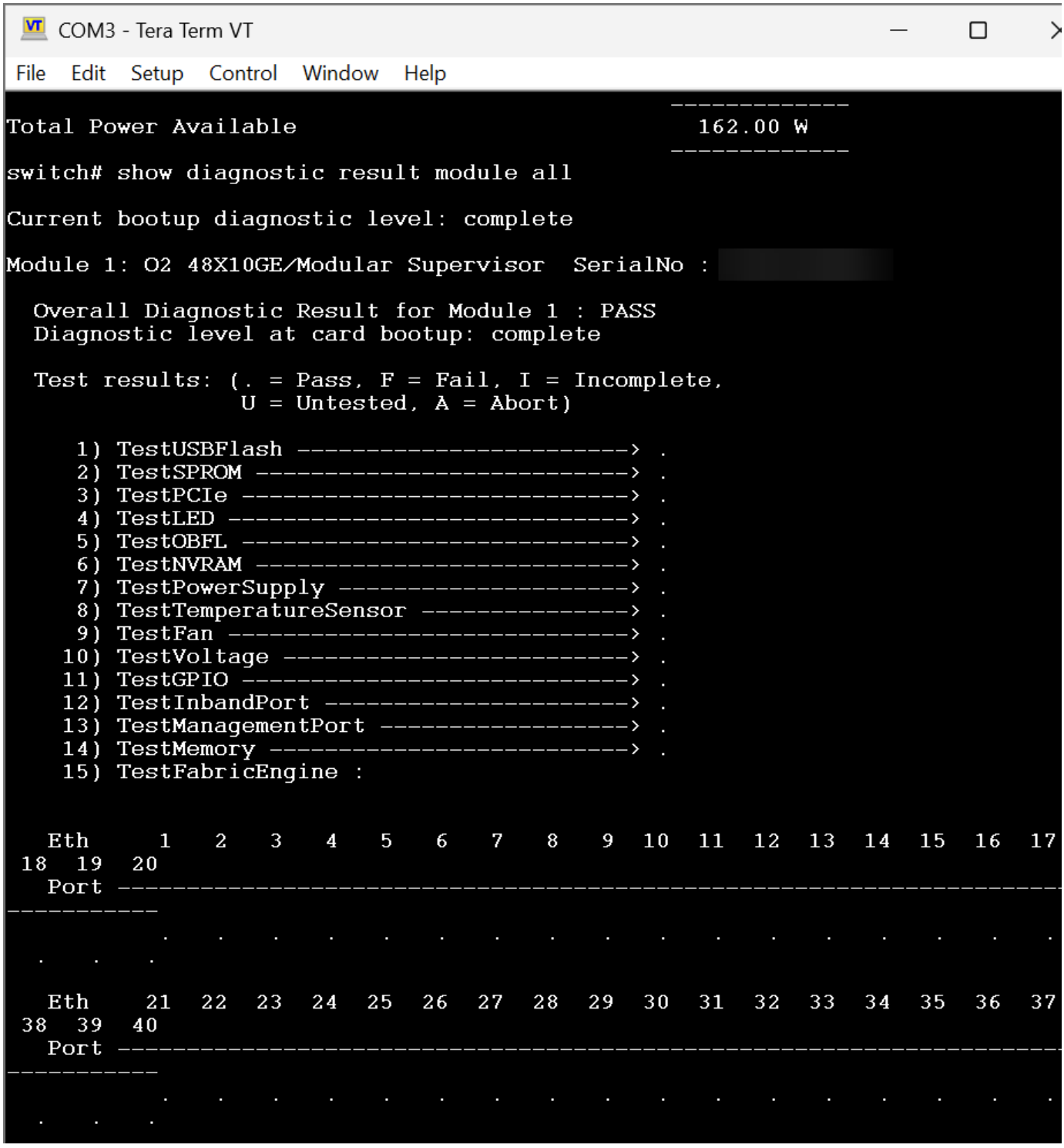
Terminal capture 10 - Recorded by Brandon Stevenson

PHASE 11

Run hardware diagnostic review

show diagnostic result module all

Displays built-in diagnostic results for the installed modules. The captured output reports PASS for module 1.



Terminal capture 11 - Recorded by Brandon Stevenson

PHASE 12

Inspect operating-system resources

show system resources

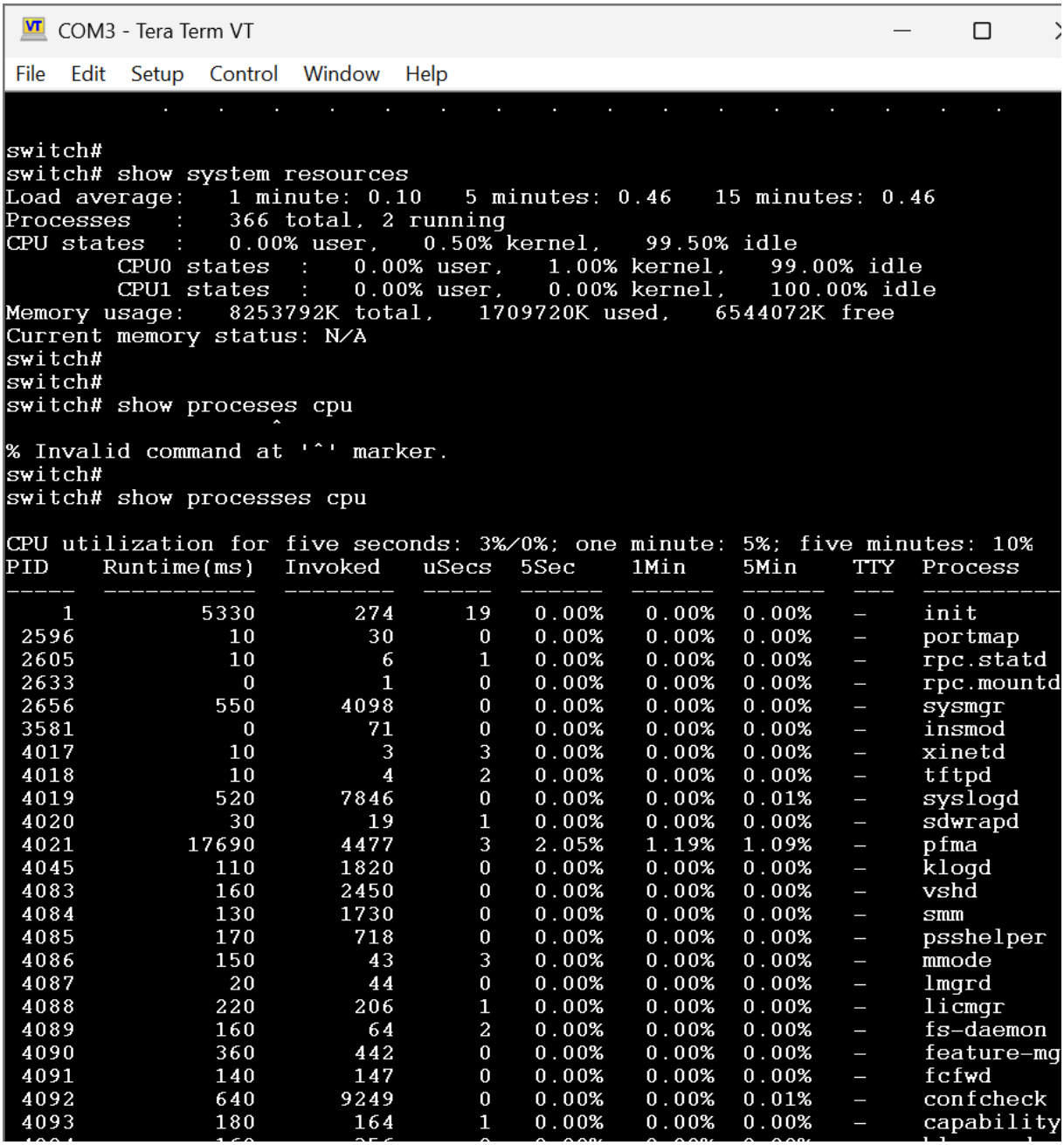
Displays current CPU utilization, system load, and memory usage.

show process cpu

The abbreviated command is rejected by this NX-OS CLI.

show processes cpu

Uses the correct syntax to display process-level CPU activity.



Terminal capture 12 - Recorded by Brandon Stevenson

PHASE 13

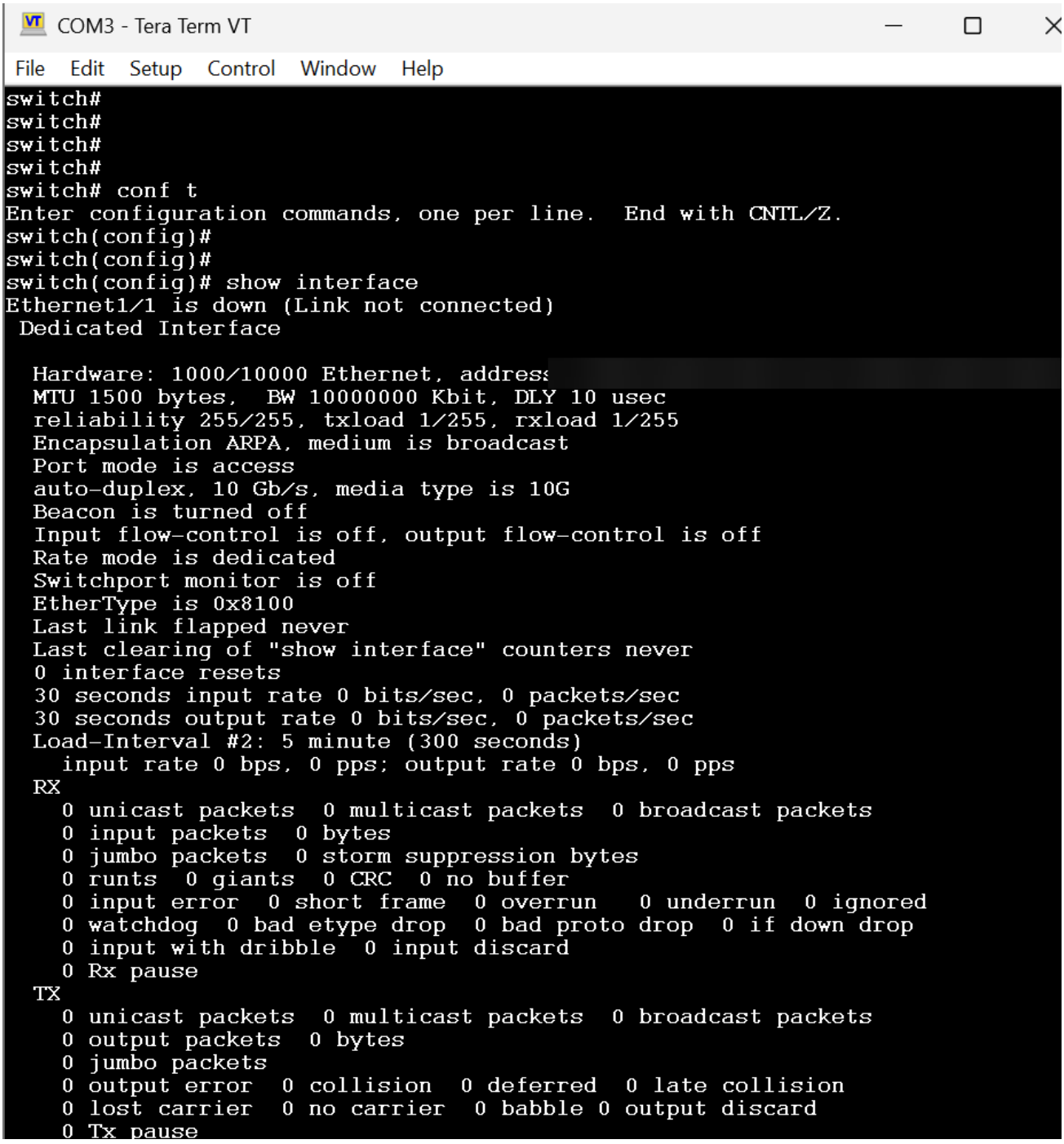
Inspect an Ethernet port

conf t / show interface

Enters configuration mode and displays detailed interface information.

Interface status and counters

Reports the selected port's link state, speed, traffic statistics, and error counters. The port is disconnected in this check.



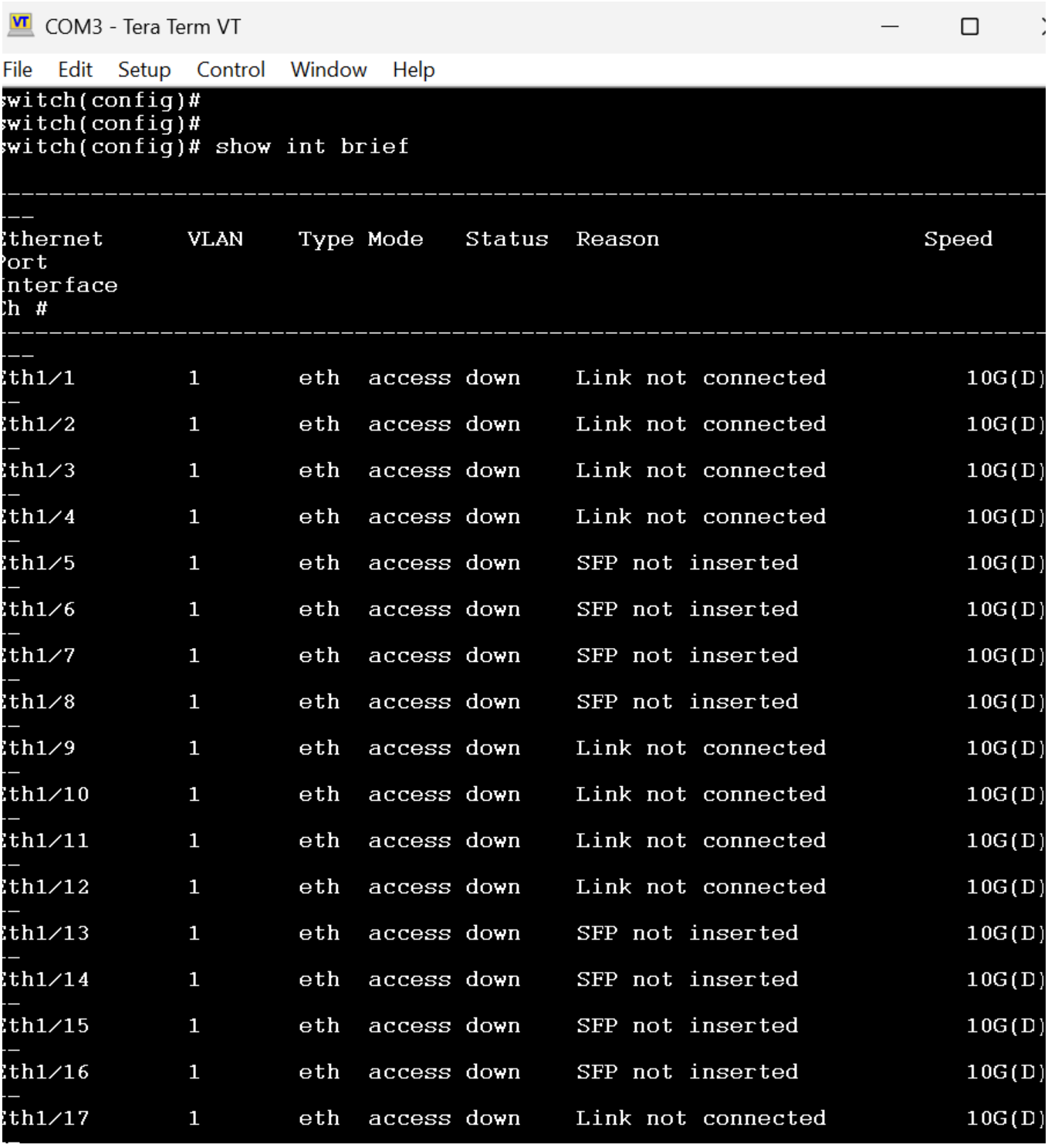
Terminal capture 13 - Recorded by Brandon Stevenson

PHASE 14

Survey Ethernet interface status

show int brief

Summarizes ports, VLAN assignments, speeds, states, and link-down reasons, including disconnected cables or missing SFPs.



Terminal capture 14 - Recorded by Brandon Stevenson

PHASE 15

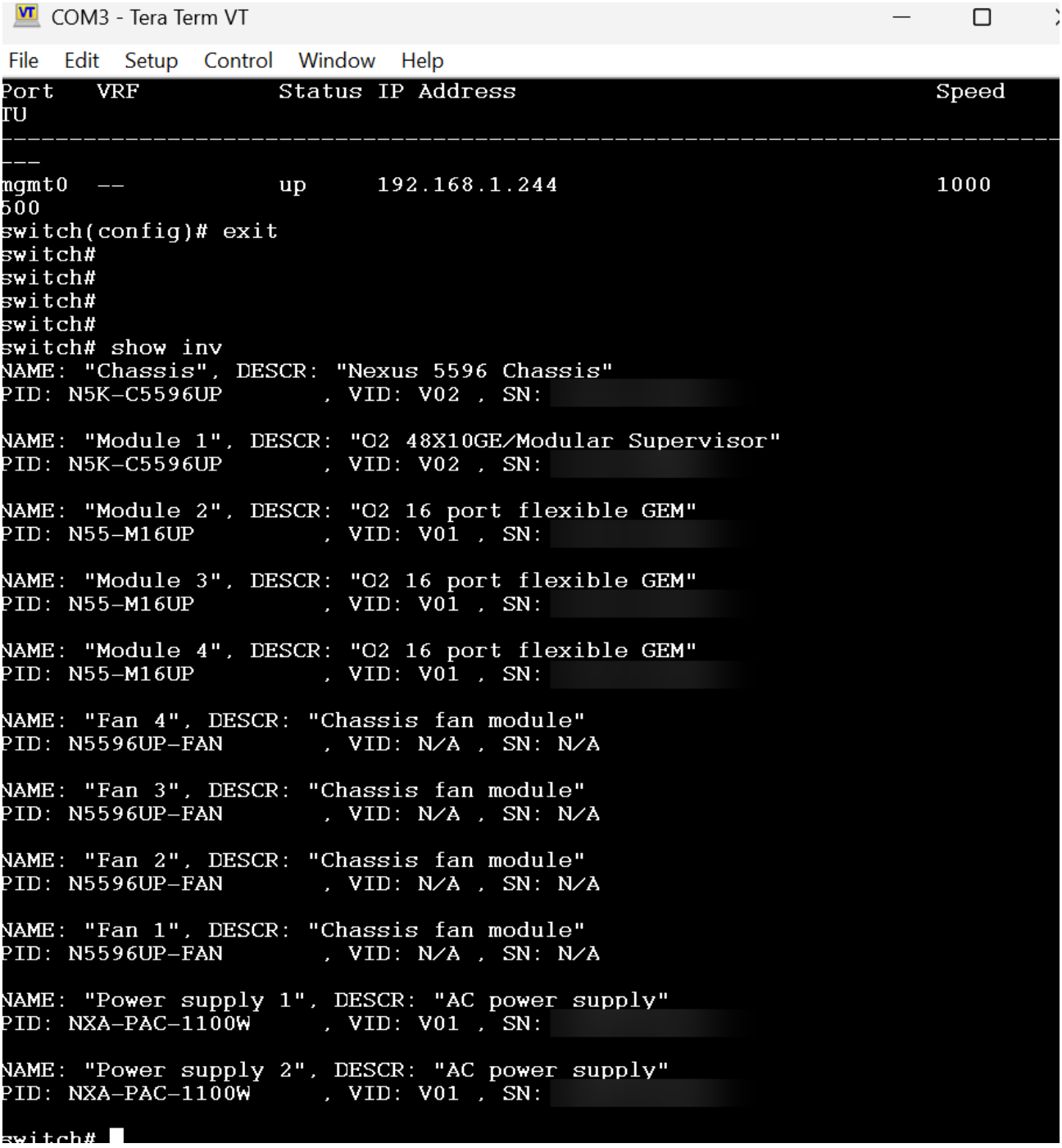
Review management connectivity and inventory

show interface brief

Displays the management port's configured IP address and status.

show inv

Identifies the chassis, supervisor, modules, fans, and power supplies. Serial-number fields are masked.



Terminal capture 15 - Recorded by Brandon Stevenson

PHASE 16

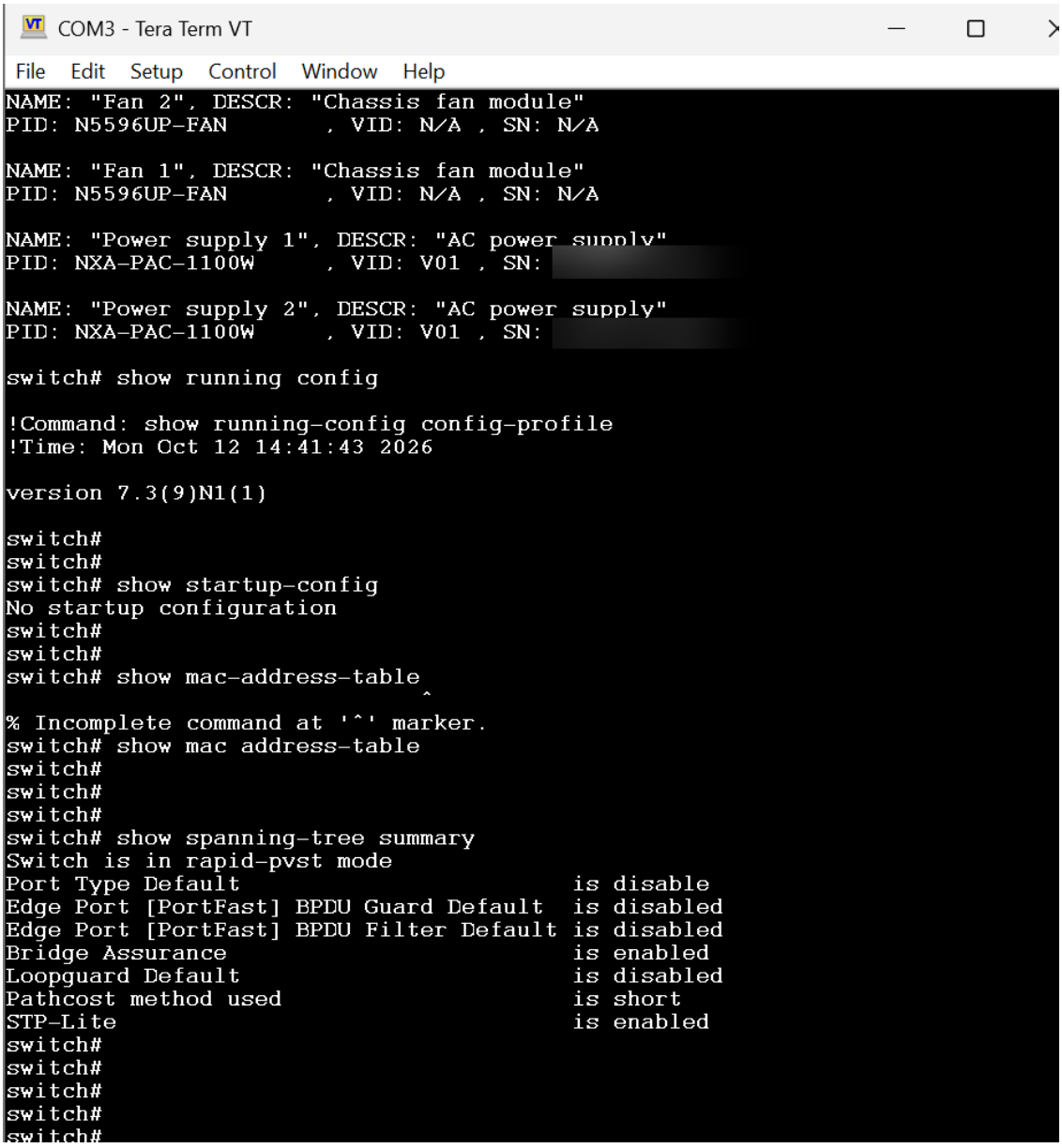
Inspect current configuration and Layer 2 state

show running-config
Displays the current system configuration.

show startup-config
Checks for saved settings; this output reports that no startup configuration is present.

show mac-address-table / show mac address-table
The first form returns an incomplete-command error. The corrected command is accepted.

show spanning-tree summary
Reports the current Spanning Tree feature settings and defaults.



Terminal capture 16 - Recorded by Brandon Stevenson

PHASE 17

Check startup diagnostic level

show diagnostic bootup level

Shows the configured startup diagnostic level. The switch reports complete.

```
switch#
switch#
switch# show diagnostic bootup level

        Current bootup diagnostic level: complete

switch#
switch#
switch#
switch#
```

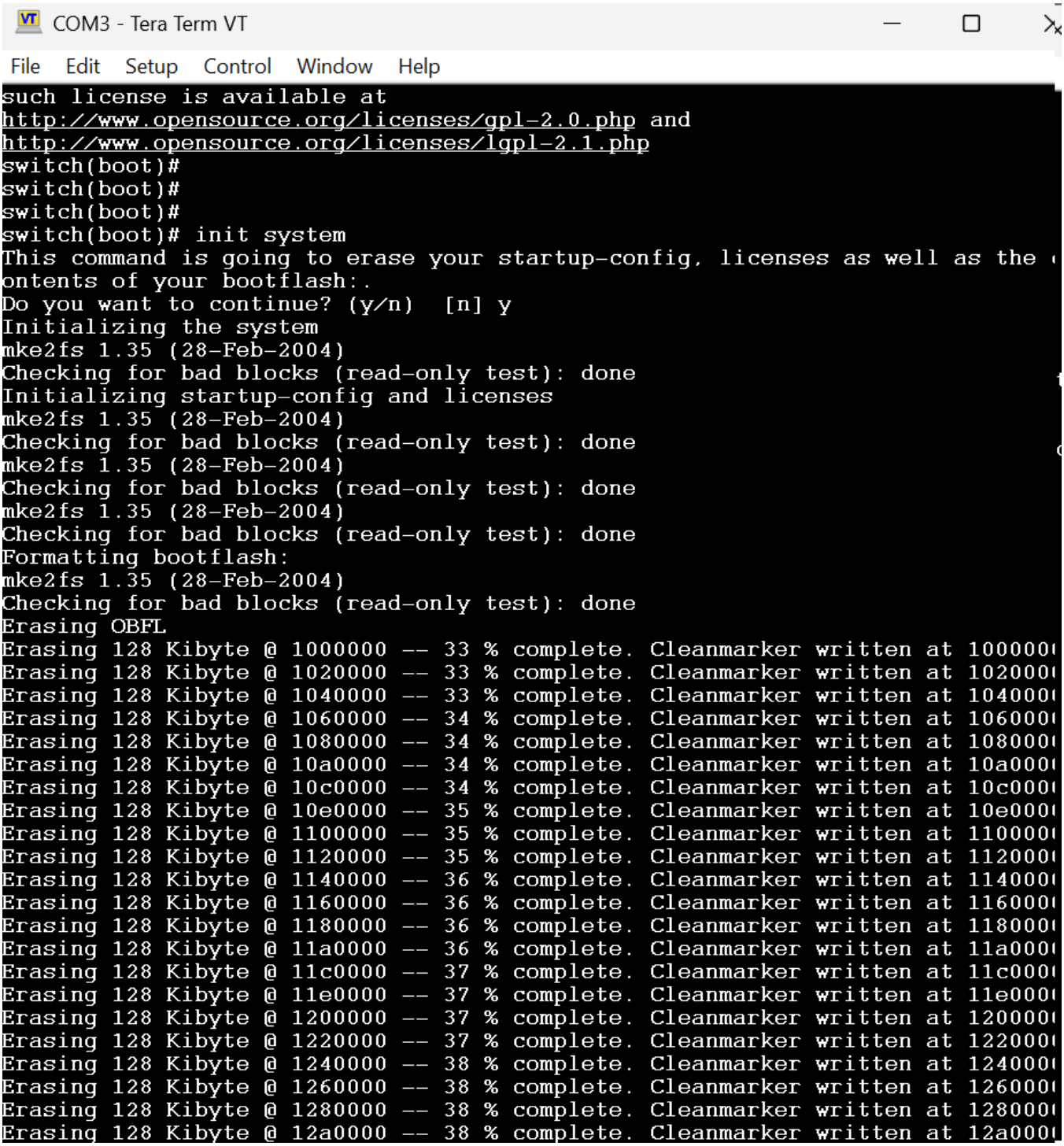
Terminal capture 17 - Recorded by Brandon Stevenson

PHASE 18

Initialize local system storage

init system

Starts system initialization, including removal of startup settings, licenses, and bootflash contents after confirmation. The console shows filesystem checks and erase progress.



Terminal capture 18 - Recorded by Brandon Stevenson

PHASE 19

Clear boot settings and inspect remaining storage

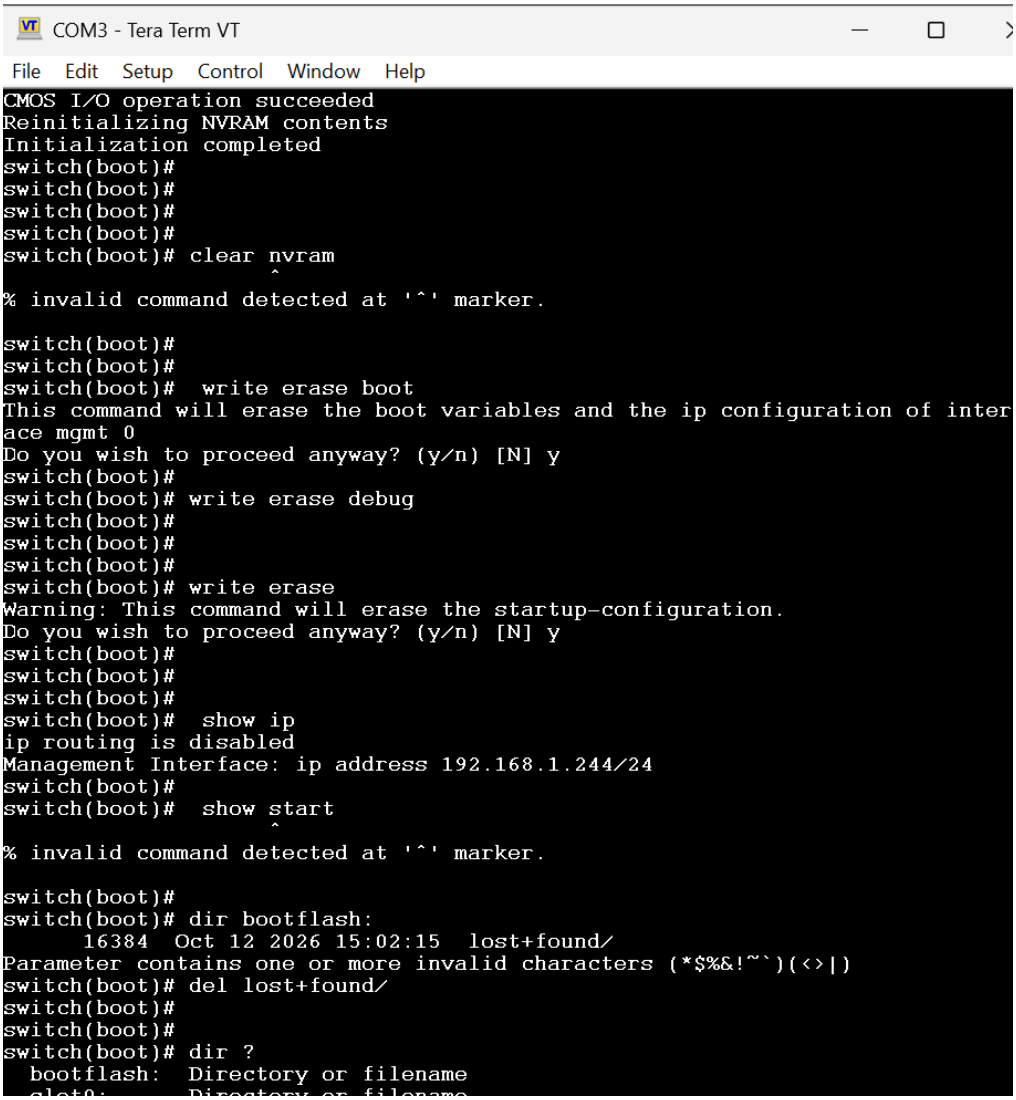
clear nvram
The command is not supported in this Nexus 5500 environment and returns an error.

write erase boot
Removes saved boot variables and boot-related management settings after confirmation.

write erase debug / write erase
Clears debug data and saved startup settings.

show ip
Displays the management-network state after clearing; the output still lists a management address.

show start / dir bootflash: / del lost+found/
Checks startup settings and bootflash contents. The shortened show command fails, while the directory listing and cleanup attempt are visible.



Terminal capture 19 - Recorded by Brandon Stevenson